Does the value of living space vary by dwelling type?

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Contributions

Wenchu Yang: Developed the analysis plan, including model comparisons, transformations, and diagnostic checks. Also worked on interpreting the preliminary results, formatting references, and integrating the team's sections.

Chenghao Li: Developed the introduction and data description, connecting the literature to the research question and explaining the data source, variable choices, and motivation for the interaction term.

Bohao Zhuang: Developed the preliminary-results section and its R code, including the diagnostic plots, dwelling-type regression lines, and coefficient table, and assessed the model's assumptions and conditions.

Leo Lu: Developed the Ethics discussion, evaluating data trustworthiness, privacy, consent, ownership, and neighborhood interpretation. Also worked on document preparation by updating the proposal PDF and section word counts.

Introduction

Among residential properties sold in Ames, Iowa, during 2006-2010, does the association between above-ground living area and sale price differ by dwelling type, after adjustment for lot area, construction quality, age, basement area, garage capacity, neighborhood, and sale year? Sale price is the recorded transaction amount in nominal US dollars. Rather than assuming a universal price-per-square-foot relationship, we estimate the adjusted sale-price difference associated with 100 additional square feet within each dwelling type. These estimates describe associations, not the causal return from enlarging a home.

Housing attributes need to be considered jointly, and past research highlights several key variables. Mathur (2019), studying King County, Washington, found that construction quality and maintenance were associated with higher prices. A medium-quality home sold for approximately 25% more than a low-quality home. This supports adjustment for quality, while the maintenance findings motivate considering condition in subsequent models. Goodman and Thibodeau (1995) analyzed nearly 8,500 Dallas single-family transactions. They found nonlinear depreciation and age-related heteroskedasticity. These findings motivate including age and checking both functional form and residual variance. Fik et al. (2003), studying the Tucson housing market, found that location and housing attributes jointly influenced prices in ways that simple location indicators did not fully capture. This motivates neighborhood adjustment, while recognizing that spatial dependence may remain.

Together, these studies support an adjusted analysis while warning against a universal price-per-square-foot rule. Our extension asks whether the living-area slope varies across five recorded dwelling types in Ames. We hypothesize that differences in layout and shared structures make a common slope too restrictive. We build on those studies by comparing what an extra 100 square feet means for the sale price of a detached home, a duplex, or a townhouse in Ames.

Multiple linear regression estimates conditional mean-price differences while adjusting for several characteristics simultaneously. A correlation or comparison of group means cannot isolate the adjusted area association or test whether slopes differ. Type-specific slopes, confidence intervals, and a joint partial F-test of the four interaction coefficients answer the question. The outcome is sufficiently continuous for this approach, subject to diagnostic assessment. An appraiser comparing historical Ames sales could use these estimates to judge whether 100 additional square feet carried the same price difference for a duplex as for a detached home. Our estimates reflect sales in Ames from 2006 to 2010. The same amount of space may be priced differently today or in another city.

Data description

Our data and codebook come from Dean De Cock's (2011a, 2011b) publication in the Journal of Statistics Education. The Ames City Assessor's Office originally recorded property characteristics and sales for valuation and assessment. To prepare the records for linear regression analysis, De Cock removed nonresidential properties, filtered out specialized tax variables, and kept only the most recent sale for repeatedly sold homes, leaving 2,930 residential properties across 82 fields.

The analysis uses 2,928 complete records, excluding two properties missing basement area or garage capacity. Sale price has a mean of $180,841, median of $160,000, standard deviation of $79,890, and range from $12,789-$755,000. Because sale price takes many dollar values, we treat it as continuous. Right-skewness in raw prices does not violate regression assumptions, which apply to conditional errors. Keeping one sale per parcel ID (PID) prevents repeated-measures dependence, though neighboring properties and shared market conditions may still induce correlated errors.

Our model includes nine predictors: living area, dwelling type, lot area, overall quality, age at sale, basement area, garage capacity, neighborhood, and sale year. Dwelling type and neighborhood are nominal; sale year is a factor. Quality is provisionally treated as a numerical 1-10 score, imposing equal increments. Living area is measured per 100 square feet and centered at 1,500 square feet; lot area is per 1,000 square feet and basement area per 100 square feet. These linear rescalings aid interpretation without affecting model assumptions.

A living-area-by-dwelling-type interaction allows distinct slopes across detached homes, two-family conversions, duplexes, and interior or end-unit townhouses. Because dwelling type is already categorical, no arbitrary numerical cutoffs are needed. Layout differences and the diverging trends in Figure 2 motivate this term.

Table 1. Complete-case summaries. Prices and areas have long upper tails. Original missing counts: basement area 1, garage capacity 1; all other selected variables 0. One age equals -1; retain for investigation. Dwelling types are imbalanced; Landmrk has one observation.

Variable	N	Mean	SD	Min	Q1	Median	Q3	Max
SalePrice	2928	180841.03	79889.90	12789.00	129500.00	160000.00	213500.00	755000.00
Living100	2928	15.00	5.06	3.34	11.26	14.42	17.42	56.42
Lot1000	2928	10.15	7.88	1.30	7.44	9.44	11.56	215.24
Quality	2928	6.10	1.41	1.00	5.00	6.00	7.00	10.00
Age	2928	36.41	30.29	-1.00	7.00	34.00	54.00	136.00
Basement100	2928	10.52	4.41	0.00	7.93	9.90	13.02	61.10
GarageCars	2928	1.77	0.76	0.00	1.00	2.00	2.00	5.00

Variable	Level	N	Percent
Building	1Fam	2423	82.75
Building	2fmCon	62	2.12
Building	Duplex	109	3.72
Building	Twnhs	101	3.45
Building	TwnhsE	233	7.96
Neighborhood	NAmes	443	15.13
Neighborhood	Blmngtn	28	0.96
Neighborhood	Blueste	10	0.34
Neighborhood	BrDale	30	1.02
Neighborhood	BrkSide	107	3.65
Neighborhood	ClearCr	44	1.50
Neighborhood	CollgCr	267	9.12
Neighborhood	Crawfor	103	3.52
Neighborhood	Edwards	194	6.63
Neighborhood	Gilbert	165	5.64
Neighborhood	Greens	8	0.27
Neighborhood	GrnHill	2	0.07
Neighborhood	IDOTRR	92	3.14
Neighborhood	Landmrk	1	0.03

Variable	Level	N	Percent
Neighborhood	MeadowV	37	1.26
Neighborhood	Mitchel	114	3.89
Neighborhood	NoRidge	71	2.42
Neighborhood	NPkVill	23	0.79
Neighborhood	NridgHt	166	5.67
Neighborhood	NWAmes	131	4.47
Neighborhood	OldTown	239	8.16
Neighborhood	Sawyer	151	5.16
Neighborhood	SawyerW	125	4.27
Neighborhood	Somerst	182	6.22
Neighborhood	StoneBr	51	1.74
Neighborhood	SWISU	48	1.64
Neighborhood	Timber	72	2.46
Neighborhood	Veenker	24	0.82
SaleYear	2006	625	21.35
SaleYear	2007	693	23.67
SaleYear	2008	621	21.21
SaleYear	2009	648	22.13
SaleYear	2010	341	11.65

Variable	Missing
SalePrice	0
Living100	0
Building	0
Lot1000	0
Quality	0
Age	0
Basement100	1
GarageCars	1
Neighborhood	0
SaleYear	0

Ethics discussion

The Ames Housing dataset uses real sales records from the Ames City Assessor's Office. It is not simulated data. De Cock (2011b) explains where the data came from, and a codebook is provided to explain the variables. This makes the dataset more trustworthy, but it does not mean the data are perfect. For example, one house has a calculated age of -1, and the overall quality score may depend on the assessor's judgment. Also, the data only include houses sold in Ames from 2006 to 2010, so the results may not apply to other places.

We also need to consider privacy and consent. The records were originally collected for property assessment, and we cannot confirm whether homeowners agreed to their use in research. Even without names, some property details might be matched with other records. We will not try to identify any owners.

For ownership, the records came from the Assessor's Office and were later published by De Cock. We will acknowledge both and use the dataset only for this project. One neighborhood has only one sale, so we will avoid publishing its parcel details or drawing conclusions about the neighborhood from that sale.

Preliminary results

The preliminary model (Table 2) regresses sale price on nine predictors plus a living area × dwelling type interaction (adjusted $R^2 = 0.839$; residual SE = \$32,073).

For detached homes, 100 additional square feet is associated with a \$4,860 higher mean price (95% CI \$4,507–\$5,212), holding other predictors fixed. The corresponding slopes are \$2,419 for two-family conversions, \$1,900 for duplexes,

$1,832 for interior townhouses and $6,042 for end-unit townhouses (Figure 2). The preliminary F-test suggests that the area slopes differ across dwelling types ($F(4, 2882) = 9.20$, $p < 0.001$). The widening residual spread in Figure 1b makes the p-value and confidence intervals provisional.

Conditional mean response: observed prices bend above the identity line at high fitted prices (Figure 1a), so this holds only partly. Conditional mean predictors: Figure 3 gives mixed evidence for the predictor condition: living and basement areas form distinct bands. Linearity is violated: residuals curve against fitted price (Figure 1b), living area (1c) and basement area (1g). Constant variance is violated: residual spread increases with fitted price (Figure 1b). Uncorrelated errors appear reasonable: residuals show no trend over sale time (Figure 1l), though nearby homes may share unmeasured influences. Normality is violated: the Q-Q plot (Figure 1m) shows heavy tails. The three largest negative residuals (rows 1499, 2181 and 2182) are partial sales above 4,000 square feet, which De Cock (2011b) flags as unlikely to reflect market value; Landmrk contains a single sale with leverage one; its standardized residual is undefined, so it is omitted from Figure 1n. No corrections are applied at this stage.

The quality coefficient ($15,672 per point) agrees in direction with Mathur (2019), the negative age coefficient (−$404 per year) is consistent with depreciation reported by Goodman and Thibodeau (1995), though our residual spread is largest for newer homes, and large neighborhood effects echo Fik et al. (2003). Unexpectedly, end-unit townhouses have a larger estimated slope than detached homes, but the difference is not significant (95% CI −$326 to $2,690). GrnHill (2 sales) and Landmrk (1) give unstable estimates.

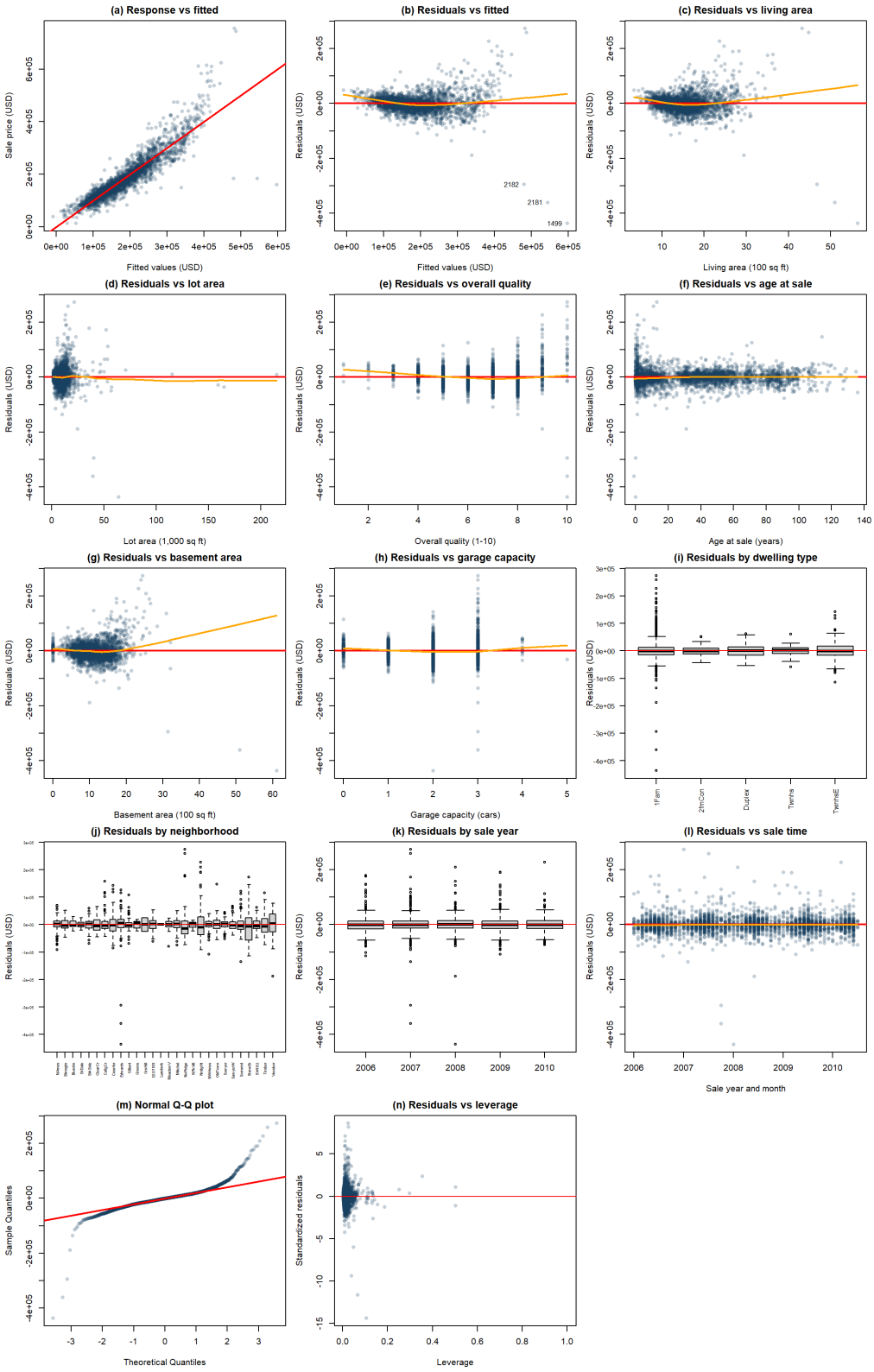


Figure 1: Residual diagnostics for the preliminary model. Orange curves are lowess smooths; labels in (b) mark the three most negative residuals.

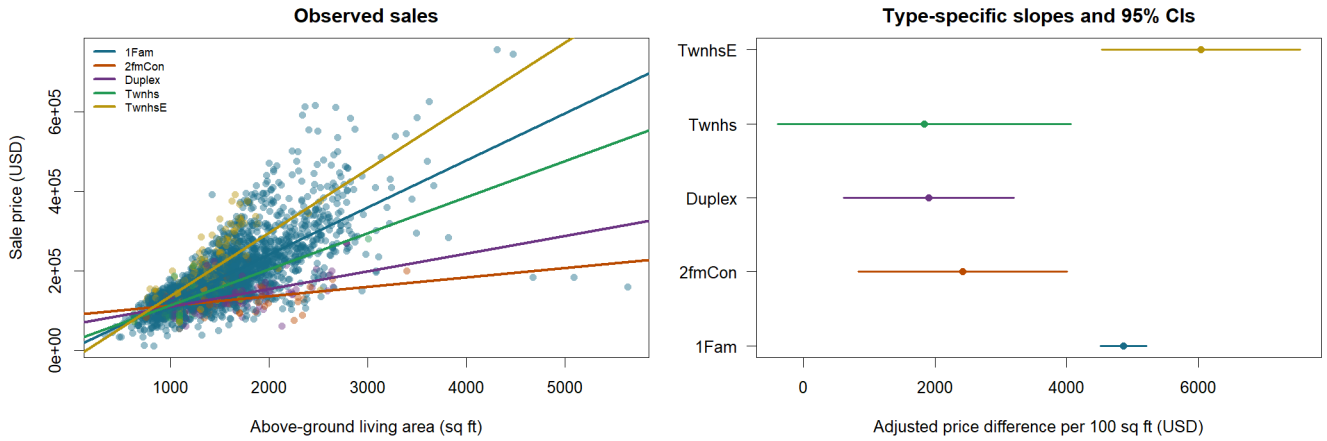


Figure 2: Observed sales with a least-squares line for each dwelling type (left) and adjusted living-area slopes with 95% CIs (right).

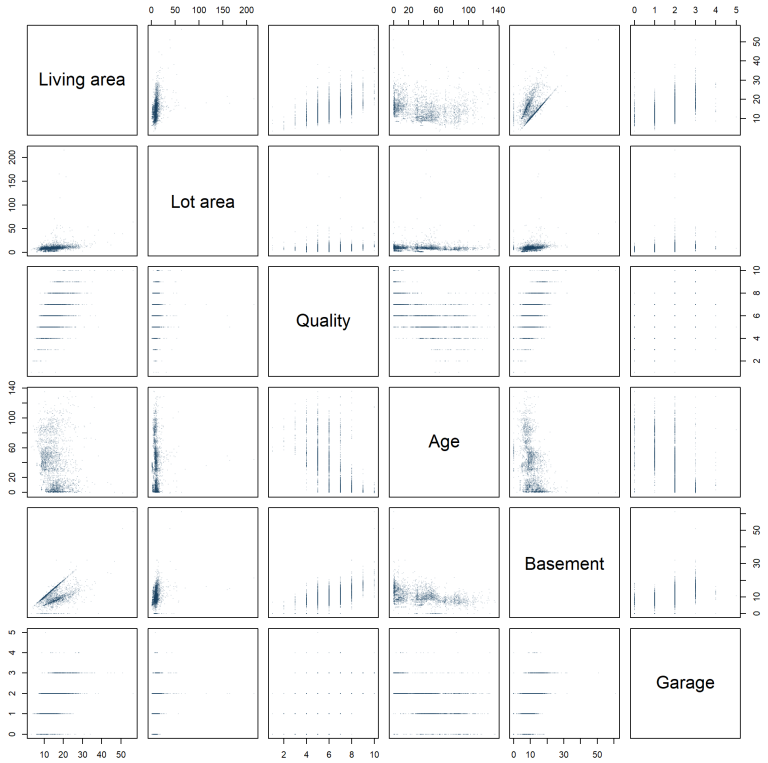


Figure 3: Pairwise scatterplots of the numerical predictors.

Table 2. Preliminary model, USD. References: detached home, North Ames, 2006. Dwelling-type contrasts are at 1,500 sq ft; interactions are slope differences per 100 sq ft. Neighborhood and sale-year effects are included but not shown.

Term	Estimate	SE	95% CI
Intercept	49,004	5,801	[37,630, 60,378]
Living area (per 100 sq ft, centred at 1,500)	4,860	180	[4,507, 5,212]
Type: two-family conversion	-9,749	4,279	[-18,139, -1,360]
Type: duplex	-21,566	3,426	[-28,285, -14,848]

Term	Estimate	SE	95% CI
Type: townhouse (interior)	-46,392	4,828	[-55,860, -36,924]
Type: townhouse (end unit)	-24,725	3,104	[-30,811, -18,639]
Lot area (per 1,000 sq ft)	529	88	[356, 702]
Overall quality (per point)	15,672	765	[14,172, 17,172]
Age at sale (per year)	-404	46	[-493, -314]
Basement area (per 100 sq ft)	2,260	182	[1,903, 2,617]
Garage capacity (per car)	10,229	1,110	[8,052, 12,406]
Living area x two-family conversion	-2,441	820	[-4,050, -833]
Living area x duplex	-2,959	674	[-4,280, -1,638]
Living area x townhouse (interior)	-3,028	1,147	[-5,277, -778]
Living area x townhouse (end unit)	1,182	769	[-326, 2,690]

Plan

We will retain living area, dwelling type, and their interaction because our question compares area slopes across types. Neighborhood and sale year will remain controls for location and market timing. Mathur (2019) motivates testing whether overall condition adds information beyond quality. Once diagnostics support inference, we will use partial F-tests to compare nested models fitted to the same sales on the same response scale. We will compare the model with and without condition. To assess simplification, we will test removing basement area and garage capacity jointly and check how much the estimated dwelling-type slopes change before dropping either control.

Figure 1b shows residual spread increasing with fitted price. We will try log price and use a Box–Cox profile to assess whether a log transformation is plausible. If curvature against living area persists (Figure 1c), we will try a squared centered-area term while retaining its linear term. After each refit, we will inspect residuals against fitted values and predictors for curvature and changing spread, the normal Q–Q plot for heavy tails, and residuals in sale order for dependence. We will compare slopes with and without the three flagged partial sales to see how sensitive our answer is to their inclusion. If we select log price, we will express area contrasts as percentage differences.

From November 9, all four members will work in parallel: Wenchu on model comparisons, Bohao on diagnostics and figures, Chenghao on literature and interpretation, and Leo on ethics, limitations, and poster design. Weekly meetings will redistribute tasks to balance workloads. Together, we will finalize analysis by November 20, integrate the poster by November 25, review it by November 27, and rehearse and record with equal speaking time by November 30. Everyone will check code, files, and access before December 2 for submission on December 3.

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